

Business Cycles and Household Formation

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To understand labor market volatility for the young, we need to understand their living arrangements

- Certain demographic groups have much larger cyclical volatility of labor market variables than others: e.g young, single people
- Understanding labor market fluctuations for these groups is key:
 - Account for very large share of overall labor market fluctuations
 - Demographic trends may lead to changes in labor market volatility
- Demonstrate that one cannot understand the labor market outcomes of these groups without understanding their living arrangements

Empirics:

Heterogeneity and volatility in household structure

For the old: little cross-sectional variation, and little changes over time

For the young: heterogeneity and volatility in living arrangements

- Differences in number and type of people with whom young people live.
Much, but not all, is due to parental coresidence
- Labor market variables differ by living arrangements
 - Young people living with their parents:
low wages, low hours, high cyclical volatility hours and wages.
 - Young people living independently:
high wages, high hours, low cyclical volatility, just like the old
- Average household size is volatile and counter-cyclical
Much, but not all, of this is due to parental coresidence.

Logic:

Unearned income and Marshallian elasticities

- Demographic groups with high volatility (the young) have very little financial wealth for smoothing shocks
- Without mechanisms for smoothing income fluctuations, Marshallian (not Frisch) elasticities drive labor market responses to wage changes
- For a given Frisch elasticity, young should have lower labor market volatility than the old, if both groups face same labor demand shocks.
- (Large) implicit transfers and lower expenses when living with old lead to increased responsiveness of labor supply to wage changes
- When labor market opportunities worsen:
 - Young people living with parents: might as well sit on parents couch
 - Young people living independently:
 - keep working to pay the bills, like the old guys, or
 - move in with an old guy and sit on their couch

Detour:

A quick history of thought

- Pervasive view in productivity driven business cycles: everything boils down to the Frisch elasticity
- Given the size of the capital stock, there is ample opportunity to smooth in a representative agent RBC model
- Many researchers noticed that young have much higher cyclical volatility of hours than old, not due to industry composition. Gomme et al. (2005), Shimer (2001)
- Early papers implicitly explained this with different Frisch's for young and old. Rios-Rull (1992,1996)

Detour:

A quick history of thought

- Jaimovich and Siu (2009) realized that this implies demographic trends have big implications for overall volatility. So it matters.
- Jaimovich, Pratt and Siu (2013): can't be due to different Frisch elasticities
Key insight: Wage volatility is also higher for young than old.
Must be labor demand differences, not labor supply differences
Technology with imperfect substitutability: higher volatilities for young than old for both wages and hours
- Problems with JPS:
 1. Qualitative argument fails with Frisch elasticities for old $< \infty$
 2. Quantitatively failure: closest they come requires Frisch for young = 7-25
 3. Living arrangements matter more than age: differences in volatility across living arrangements among young are larger than young vs old

Model:

Quantifying importance of coresidence

- Simple model with minimal deviations from RBC and JPS
- Estimate based on the relevant data:
 - Relative hours and wage variances of old, young alone, young together
 - Measures of cyclical of living arrangements
- Match relative variances with reasonable Frisch for old
(Frisch for young is not well-defined, action from Marshallian elasticity)
- **Counterfactual experiments:** How much of additional hours volatility for young is due to the coresidence effect vs the volatility in coresidence dynamics vs technology
All matter but first effect is far more important
- **Useful metric:** How big should one set the Frisch in an RA model to implicitly capture the Marshallian effects of the young?
Frisch = 0.7 for old, implies a RA Frisch = 1.3

Cyclical volatility of living arrangements

Definition of young:	18-30	Never married	18-30 & never married
Fraction young	32%	26%	19%
Fraction young live with old	50%	52%	67%
St Dev log fraction live with old	0.83%	0.80%	0.66%
Corr: log fraction young live with old, log hours of 18-65	-0.48	-0.33	-0.37

- Source: CPS Basic monthly surveys, 1979:Q1-2010:Q3
- Sample: Civilians aged 18-65 not in school

Living arrangements and the labor market

Definition of young:		18-30	Never married	18-30 & never married
Av. hours:	All young	0.91	0.89	0.84
	Young alone	1.06	1.03	1.06
	Young with old	0.75	0.75	0.73
Av. wages:	All young	0.57	0.62	0.51
	Young alone	0.72	0.82	0.72
	Young with old	0.43	0.43	0.40
Var log hours:	All young	2.74	2.69	3.36
	Young alone	1.78	1.77	1.84
	Young with old	4.01	4.22	4.63
Var log wages:	All young	2.08	1.48	2.22
	Young alone	2.21	1.63	2.23
	Young with old	2.57	1.99	2.70

- Table reports statistics relative to those for the old group
- Source: CPS Basic monthly surveys, 1979:Q1-2010:Q3, March supplement for wage data
- Sample: Civilians aged 18-65 not in school

Demographics

- Continuum of agents of measure 1
- Old agents (μ)
 - Live in groups of size γ : there are μ/γ households
 - Can be invaded by a young agent, but only after choice of consumption and hours has been made
- Young agents ($1-\mu$)
 - Can join (invade) a stable household after observing state and realization of an IID idiosyncratic shock $\eta \sim F(\eta; \lambda)$
 - Fraction of young who invade an old household = x
- At any point there are three types of agents:
 1. Old
 2. Young alone
 3. Young together (with old)

Old agents

$$u(c^o, h^o, x) = \left[1 - \frac{x(1-\mu)\gamma}{\mu}\right] \left[\log \frac{c^o}{\zeta^{oo}} - \psi^o \frac{(h^o)^{1+\frac{1}{\nu^o}}}{1+\frac{1}{\nu^o}}\right] + \frac{x(1-\mu)\gamma}{\mu} \left[\log \left(\frac{c^o}{\zeta^{oo} + \zeta^o}\right) - \psi^o \frac{(h^o)^{1+\frac{1}{\nu^o}}}{1+\frac{1}{\nu^o}}\right]$$

- Economies of scale among old: ζ^{oo}
- (Meager) economies of scale from young: ζ^o
- Discount future at rate β

Young agents

- Young alone

$$u(c^{yA}, h^{yA}) = \frac{(c^{yA})^{1-\sigma}}{1-\sigma} - \psi^y \frac{(h^{yA})^{1+\frac{1}{\nu^y}}}{1+\frac{1}{\nu^y}}$$

- Young together with old

$$\begin{aligned} u(c^{yT}, h^{yT}, \eta) &= \frac{(c^{yT} + \zeta^y)^{1-\sigma}}{1-\sigma} - \psi^y \frac{(h^{yT})^{1+\frac{1}{\nu^y}}}{1+\frac{1}{\nu^y}} - \eta \\ &\quad + [(1 + [x_{t-1} - x_{SS}])^{\gamma_2} - 1]. \end{aligned}$$

- ζ^y : economies of scale
- η : dislike of living with old
- Final term is an externality: strength of social stigma

Budget constraints

- Young are less productive than old ($\varepsilon^y < 1$) and are hand-to-mouth

$$c^{yA} = \varepsilon^y w^y h^{yA}, \quad c^{yT} = \varepsilon^y w^y h^{yT}$$

- Old have a standard budget constraint

$$c^o + a' = w^o h^o + a(1 + r)$$

Technology

- Nested CES with capital-experience complementarity (JPS)

$$F(z, K, N^y, N^o) = \left[\mu_F (zN^y)^\sigma + (1 - \mu_F) (\lambda_F K^\rho + (1 - \lambda_F) (A_t N^o)^\rho)^{\sigma/\rho} \right]^{1/\sigma}$$

where N^y and N^o are labor inputs of young and old

- Technology allows us to jointly match higher hours and wage volatility of young than old

Aggregation

- Structure is on top of a standard RBC model

$$C + K' = F(z, K, N^y, N^o) + (1 - \delta)K$$

where z is an AR(1) productivity shock

$$C = \frac{\mu}{\gamma} c^o + (1 - \mu) [x c^{yT} + (1 - x) c^{yA}]$$

$$N^o = \frac{\mu}{\gamma} h^o$$

$$N^y = (1 - \mu) \epsilon^y [x h^{yT} + (1 - x) h^{yA}]$$

$$N = N^o + N^y,$$

$$H = \frac{\mu}{\gamma} h^o + (1 - \mu) [x h^{yT} + (1 - x) h^{yA}]$$

- Capital is owned by old, so wealth is total capital: $K = \frac{\mu}{\gamma} a$

Calibration: Technological parameters

Parameter	Description	Parameter	Target variable	Target
<u>Parameters that can be set without solving the model</u>				
ρ	Elasticity parameter	.20	Micro estimates JPS (2012)	.20
σ	Elasticity parameter	.66	Micro estimates JPS (2012)	.66
<u>Parameters that require solving for the steady state</u>				
A	Normalization	3.39	Y	1.0
δ	Depreciation rate	.02	I/Y	.24
μ_F	Weight of young	.40	Capital Share	.37
λ_F	Weight of capital	.28	Share of olds' income in total	.52
ε^y	Labor efficiency of young	.61	Relative wages	.57
<u>Parameters that require solving the full stochastic equilibrium</u>				
ρ_z	AR(1) prod shocks	.94	Autocorr AR(1) RA Solow Res*	.94
σ_z	St Dev productivity shocks	1.01%	Var Solow Res*	3.19

Calibration: Preference parameters for old

Parameter	Description	Target variable	Target	Parameter
<u>Parameters that can be set without solving the model</u>				
μ	Fraction of old	Measurement	.68	.68
γ	Old household size	Direct Measurement	1.80	1.80
β	Discount rate	Interest rate	.04	.99
ζ^{oo}	Ec of scale for old	OECD	1.70	1.70
ζ^o	Additional Ec of scale	OECD	.50	.50
ν^o	Frish elast of old	Measurement	.72	.72
<u>Parameters that require solving for the steady state</u>				
ψ^o	Weight of hours of old	Hours in Old Hholds	.50	4.16

Calibration: preference parameters for young

Parameter	Description	Target	Target variable	Model counterpart	Parameter value
All Parameters require solving the full stochastic equilibrium					
ψ^y	Weight of hours of young	Hours of young together	.21	.21	7.78
λ^1	Shape par of Gamma dist	% of young with old	50.2	50.2	.154
σ^y	Risk Aversion of young	Hours of young alone	.29	.29	.25
λ^2	Scale par of Gamma dist	Moment $M(h^{yT}, h^{yA}, x)$	.11	.11	1.52
ν^y	Curvature in utility for young	$Var(hy)/Var(ho)$	2.74	2.92	.85
ζ^y	Ec of scale of young	$Var(hyA)/Var(hyT)$	0.44	0.44	1.96
γ_2	Curvature in externality	$Corr(x, h)$	-.47	-.47	3.69

Notes: Variances are of logged and HP-filtered series.

Model fit: not recalibrated

	Data	Baseline model	Baseline model ($x = x_{ss}$)	Baseline model ($x = 0$)
$var(h^y)/var(h^o)$	2.74	2.93	2.64	1.93
$var(h^{yA})/var(h^o)$	1.78	1.78	1.80	1.93
$var(h^{yT})/var(h^o)$	4.01	4.01	4.07	-
$var(w^y)/var(w^o)$	2.08	1.66	1.69	1.81
$var(w^{yA})/var(w^o)$	2.21	-	-	-
$var(w^{yT})/var(w^o)$	2.57	-	-	-

Model fit: recalibrated economies

	Data	Baseline model	Baseline model ($x = x_{ss}$)	Baseline model ($x = 0$)*
$var(h^y)/var(h^o)$	2.74	2.93	2.59	2.73
$var(h^{yA})/var(h^o)$	1.78	1.78	1.78	2.73
$var(h^{yT})/var(h^o)$	4.01	4.01	4.01	-
$var(w^y)/var(w^o)$	2.08	1.66	1.70	1.68
$var(w^{yA})/var(w^o)$	2.21	-	-	-
$var(w^{yT})/var(w^o)$	2.57	-	-	-

A metric for the effect on aggregate hours

Frisch elasticity for the stable:	$var(h)$	Implied Frisch in RA RBC	Proportional Increase
$\nu = 0.72$ (Baseline)	.240	1.31	82%
$\nu = 0.4$	.091	.66	64%
$\nu = 0.7$	.237	1.30	86%
$\nu = 1.0$	.354	1.83	83%
$\nu = 1.3$	.510	2.63	102%
$\nu = 1.6$	.678	3.70	131%

Alternative calibrations

	Baseline model	Never married	Never married 18-30	Implicit definition	Target $\frac{var(x)}{var(h^o)}$	Economy w/o ext.	No ext. $+\lambda^1$ iid
$var(h^y)/var(h^o)$	2.93	2.89	3.62	2.89	2.80	2.92	2.91
$var(h^{yA})/var(h^o)$	1.78	1.77	1.84	1.78	1.78	1.78	1.78
$var(h^{yT})/var(h^o)$	4.01	4.22	4.63	4.01	4.01	4.01	4.01
$var(w^y)/var(w^o)$	1.66	1.63	1.49	1.67	1.67	1.66	1.66
$var(w^{yA})/var(w^o)$	-	-	-	-	-	-	-
$var(w^{yT})/var(w^o)$	-	-	-	-	-	-	-

Implicit transfers, earnings and consumption

	Baseline model
Earnings old (e^o)	1.68
Earnings young together (e^{yT})	.33
Implicit transfer (ζ^y)	1.96
Ratio ζ^y/e^o	1.17
Ratio ζ^y/e^{yT}	6.05
Consumption old	1.68
Consumption young alone	.45
Consumption young together	.33

Contributions

- Simultaneously match all relative hours and wage means and variances of old, young alone and young together
- General insights transcend focus on young and living arrangements
 - Introducing coresidence
 - Emphasizing Marshallian rather than Frisch elasticities
 - Unearned income takes many forms
- Our results suggest that both labor supply and labor demand differences are important (we retain same technology as in JPS)
- This has consequences for the long-run trends since it is coresidence rates not the number of young that matter...!!!